Maths Review for Machine Learning and Data Science
Topic 1: Functions and Notation

Jacey Ligthart-Smith
Senior Learning Skills Officer (Statistics and Data)
Academic Skills Centre (ASC)

Semester One, 2026

The presentation slides from today's session will be made available on the CITS5508 and Master of Data Science Community LMS pages.

Contents

1. Introduction Go
2. Notation Go
   1. Summation and product Go
   2. Sets Go
3. Functions Go
   1. What is a function? Go
   2. Example: Linear functions Go
   3. Example: Square root Go
   4. Domain and Codomain Go
   5. Inverse Go
   6. Example: Logarithms and exponentials Go
   7. Piecewise functions Go
   8. Example: The sign function Go
4. Useful resources Go
5. UWA's Academic Skills Centre Go

Maths Review for Machine Learning and Data Science:

1. **Functions and Notation**
2. Linear Algebra
   - Vectors
   - Matrices
   - Vectorization
3. Calculus
   - Differentiation
   - Partial derivatives
   - Matrix calculus
4. Optimisation
   - Minima and maxima
   - Argmin and Argmax
   - Convex functions

Introduction

A few important notes:

- These workshops will review the fundamentals of each topics. Suggestions for further reading will be given for those wishing to review or learn more advanced content.
- Based on the EOI responses, we will not be covering basic algebra. You are welcome to attend the Academic Skills Centre online workshop on Algebra in Week 2. If you register for the workshop, you will receive the slides whether or not you attend. If you have missed the workshop, you can email me to request a copy of the slides.
- If you have any questions during the workshop, please raise your hand and ask! If you would like to discuss any of these topics in more detail, one-on-one, there will be information at the end about accessing our free drop-in and consultation services.

Notation: Summation (Sigma)

Summation notation allows us to succictly represent a sum of many terms.

For example:

$$\sum_{i=1}^{5} i = 1 + 2 + 3 + 4 + 5$$

Here, our index i represents the number to be added.

However, it could instead represent the index in data set:

Let $\mathbf{x} \leftarrow [x_1, x_2, x_3, x_4, x_5] = [2, 7, 3, 4, 1]$, then:

$$\sum_{i=1}^{5} x_i = x_1 + x_2 + x_3 + x_4 + x_5 = 2 + 7 + 3 + 4 + 1$$

Notation: Summation (Sigma)

You can think of the summation as a loop. For example:

$$\sum_{i=1}^{n} i = 1 + 2 + ... + n$$

Can be computed (in Python) using:

```
sum = 0
for i in range(1, n+1):
    sum += i
```

Notation: Summation (Sigma)

A few useful summation rules:

The sum of a constant:

$$\sum_{i=1}^{n} c = nc$$

Constant factors:

$$\sum_{i=1}^{n} cx_i = c\sum_{i=1}^{n} x_i$$

Addition:

$$\sum_{i=1}^{n} x_i + y_i = \sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i$$

Subtraction:

$$\sum_{i=1}^{n} x_i - y_i = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} y_i$$

For a more in-depth explanation of the Sigma notation and why each of these rules are true, you can attend the Academic Skills Centre workshop in Week 3.

Notation: Product (Pi)

Similarly, product notation allows us to succictly represent a product of many terms.

For example:

$$\prod_{i=1}^{5} i = 1 \times 2 \times 3 \times 4 \times 5$$

Here, our index i represents the number to be multiplied. However, it could instead represent some data.

Let $\mathbf{x} \leftarrow [x_1, x_2, x_3, x_4, x_5] = [2, 7, 3, 4, 1]$, then:

$$\prod_{i=1}^{5} x_i = x_1 \times x_2 \times x_3 \times x_4 \times x_5 = 2 \times 7 \times 3 \times 4 \times 1$$

Notation: Product (Pi)

A few useful product rules:

The sum of a constant:

$$\prod_{i=1}^{n} c = c^n$$

Constant factors:

$$\prod_{i=1}^{n} cx_i = c^n \sum_{i=1}^{n} x_i$$

Multiplication:

$$\prod_{i=1}^{n} x_i \cdot y_i = \prod_{i=1}^{n} x_i \cdot \prod_{i=1}^{n} y_i$$

Division:

$$\prod_{i=1}^{n} \frac{x_i}{y_i} = \frac{\prod_{i=1}^{n} x_i}{\prod_{i=1}^{n} y_i}$$

Notation: Sets

In maths, we use *curly braces* $\{\}$ to denote *sets*.

A set is a collection of objects. We are interested only in whether something is an element of the set or not, so order and repititon do not affect equality of sets:

$$\{1, 2, 3\} = \{3, 2, 1\} = \{1, 2, 3, 3, 2, 1\}$$

We usually denote sets using capital letters, for example:

$$A = \{1, 2, 3\}$$

Notation: Sets

Set relations:

- $\in$: An element of, e.g. $x \in \{x, y, z\}$
- $\subset$: A subset of, e.g. $\{x, z\} \subset \{x, y, z\}$
- $\cup$: The union, i.e. the elements of either set, e.g.

$$\{x, z\} \cup \{y, z\} = \{x, y, z\}$$

- $\cap$: The set intersection, i.e. the elements in both sets, e.g.

$$\{x, z\} \cap \{y, z\} = \{z\}$$

- $\setminus$: The set difference, i.e. elements of the left but not the right set, e.g.

$$\{x, y, z\} \setminus \{x, y\} = \{z\}$$

Notation: Sets

A few important sets:

- $\emptyset$: The empty set, $\{\}$

- $\mathbb{N}$: The set of all natural numbers, $\{[0,]\,1, 2, 3, ...\}$
- $\mathbb{Z}$: The set of all integers, $\{... - 3, -2, -1, 0, 1, 2, 3, ...\}$
- $\mathbb{Q}$: The set of all rational numbers $\left\{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\right\}$
- $\mathbb{R}$: The set of all real numbers, that is, all the numbers along the continuous number line

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

What is a function?

Much like in programming, mathematical *functions* have *inputs* and *outputs*.

We often denote functions using the letters f, g, h. For example:

$$f(x) = x^2$$

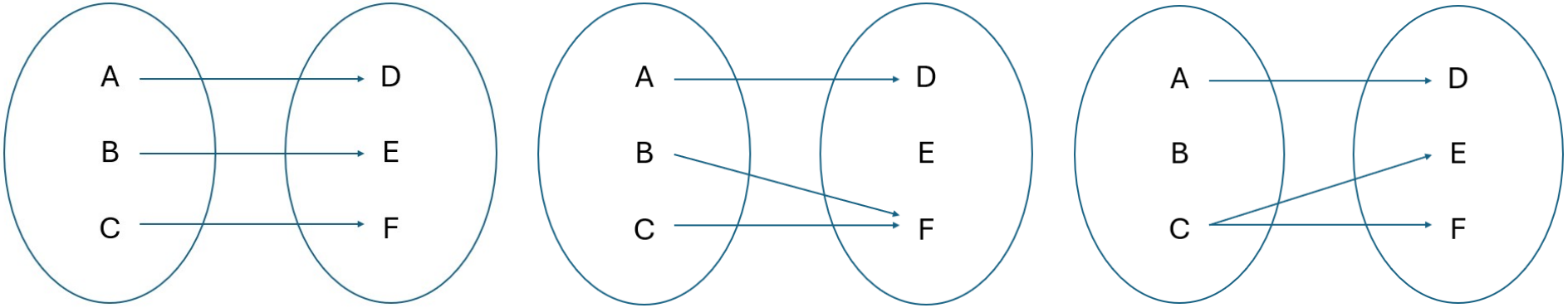
takes the input (or *variable*) x and *maps* it to the output (or *value*) x^2.

What is a function?

Importantly, a mathematical function must only have one output for every input.

Lets say our set of possible inputs to is {A, B, C} and our set of possible outputs is {D, E, F}.

Which of the following relations depicts a function?

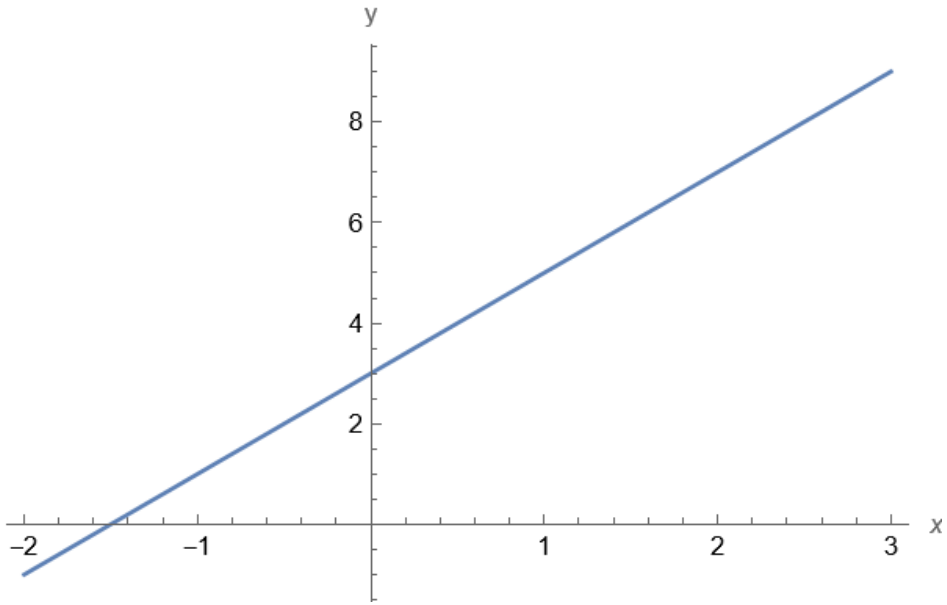


The first two! A function can map two inputs to the same output, but not one input to two different outputs.

Example: Linear functions

You are probably familiar with the formula $y = mx + c$ to denote a line in two dimensions, with gradient m and y-intercept c.

For example, $y = 2x + 3$ represents a line on the Euclidean (x, y) plane that crosses the y-intercept at 3 and increases in height by 2 for each increase of 1 in the x-axis.



Example: Linear functions

You are probably familiar with the formula $y = mx + c$ to denote a line in two dimensions, with gradient m and y-intercept c.

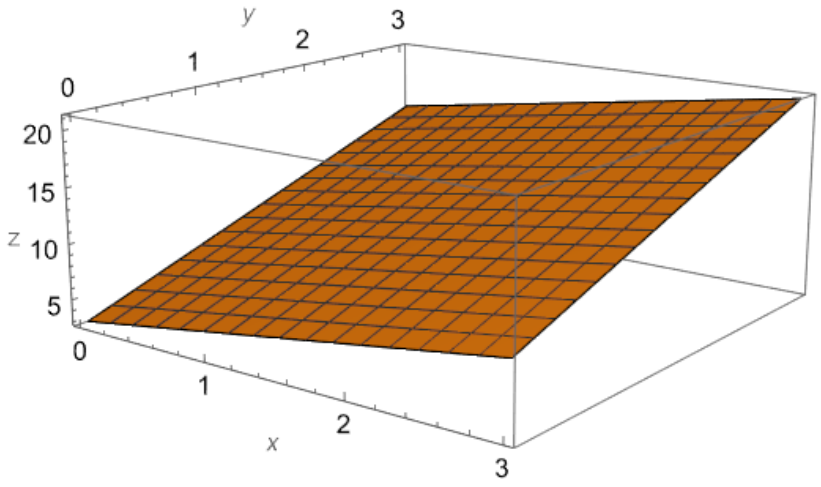
You might also see this written as:

- $y = ax + b$
- $f(x) = ax + b$
- $\hat{y} = \theta_0 + \theta_1 x$

These are all different ways to represent the same function.

Example: Linear functions

You can also have linear functions in more than two dimensions. For example, $z = 2x + 4y + 3$:

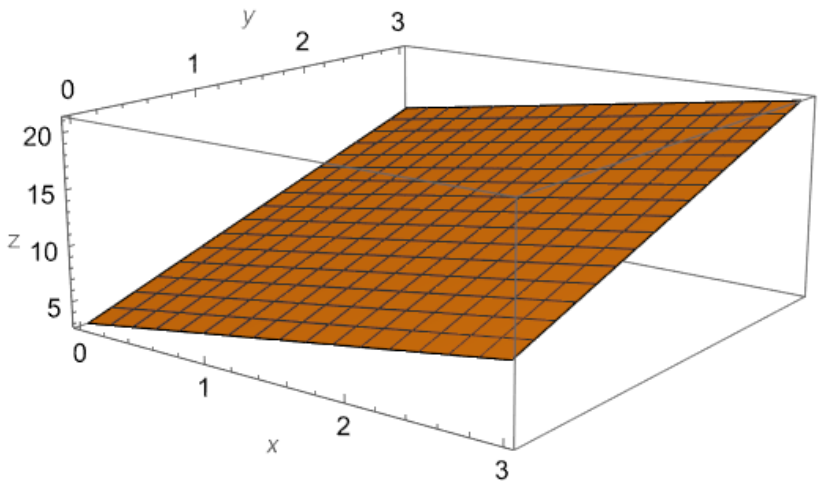


You could also write this as:

- $f(x, y) = 2x + 4y + 3$
- $\hat{y} = 3 + 2x_1 + 4x_2$

Example: Linear functions

You can also have linear functions in more than two dimensions. For example, $z = 2x + 4y + 3$:



You can extend beyond 3 dimensions (i.e. beyond 2 variables), but it becomes impossible to visualise. When you learn about linear regression, you might see the formula

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + ... + \theta_n x_n,$$

which describes a linear function of n variables.

Example: Square root

The *square root* of a number x is the set of numbers that, when squared (multiplied by themselves) give x.

For example, the square roots of 9 are 3 and -3.

Is the square root a function?

No!
A single input can produce more than one output!

The square root *function*, denoted using the *radical symbol*, $\sqrt{}$ is defined as the *non-negative* square root.

That is, $\sqrt{9} = 3$.

Functions: Domain, Codomain, and Range

The *domain* of a function is the set of possible inputs, and the *codomain* is the set of possible outputs.

For example, the domain and the codomain of the $f(x) = x^2$ might the the set of real numbers, $\mathbb{R}$.

That is, $f : \mathbb{R} \to \mathbb{R}$.

The domain and codomain are part of the function definition, so changing them can change the function.

The $range$, however, is the set of actual possible outputs of a function based on the codomain. The range of our function f is $[0, \infty)$.

Functions: Domain, Codomain, and Range

The difference between the codomain and the range is more obvious for some functions than others.

For example, if we define $g(n) = n^2$ as a function squaring only integers, we have a mapping from the integers to the natural numbers (including 0), that is,

$$g : \mathbb{Z} \to \mathbb{N}.$$

However, not every element of $\mathbb{N}$ is a square. The range of our function would be

$$\{0, 1, 4, 9, 16, 25, ...\}.$$

What are the domain and codomain of linear functions?

In two dimensions, the domain is usually $\mathbb{R}^2$ and the codomain $\mathbb{R}$.

$\mathbb{R}^2$ is the set of *Cartesian coordinates*, or pairs of real numbers.

That is, $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$.

We can define our n-dimensional linear function,

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + ... + \theta_n x_n,$$

on domain and codomain $\hat{y} : \mathbb{R}^n \to \mathbb{R}$

Functions: Inverse

The *inverse* of a function reverses it's relation, mapping from the codomain to the domain.

We've already seen an inverse (sort of).

If we define $f : [0, \infty) \to \mathbb{R}$ as

$$f(x) = x^2,$$

then the inverse, $f^{-1} : \mathbb{R} \to [0, \infty)$ is given by

$$f(x) = \sqrt{x}.$$

Why did we need to constrain the domain of f?

Note: We could also write the co-domain as $[0, \infty)$, but that is less important to our definition.

Functions: Inverse

The *inverse* of a function reverses it's relation, mapping from the codomain to the domain.

We've already seen an inverse (sort of).

If we define $f : [0, \infty) \to \mathbb{R}$ as

$$f(x) = x^2,$$

then the inverse, $f^{-1} : \mathbb{R} \to [0, \infty)$ is given by

$$f^{-1}(x) = \sqrt{x}.$$

This goes both ways, so $\left(f^{-1}\right)^{-1} = f$.

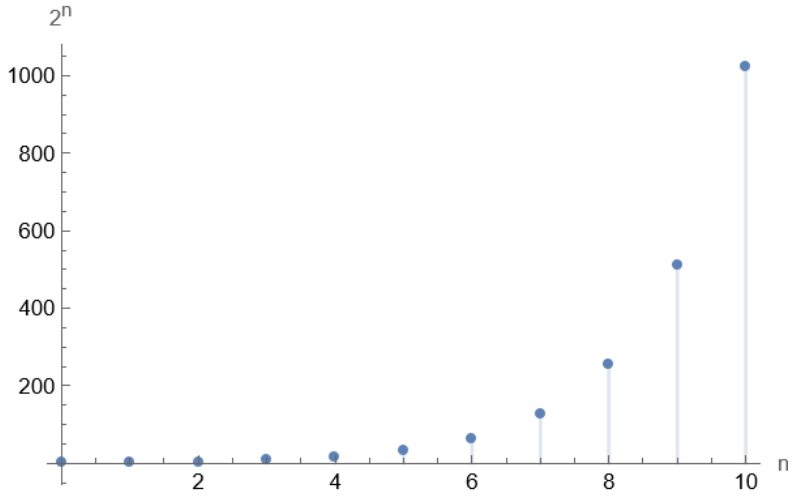
Example: Logarithms and exponentials

An *exponential* function is one where some number is taken to the power of a variable.

For example, let $f(n) = 2^n$ defines a function $f : \mathbb{N} \to \mathbb{N}$ such that we multiply 2 by itself n times.

So, $f(1) = 2 \times 1 = 2$, $f(2) = 2 \times 2 = 4$, $f(3) = 2 \times 2 \times 2 = 8$, and so on.

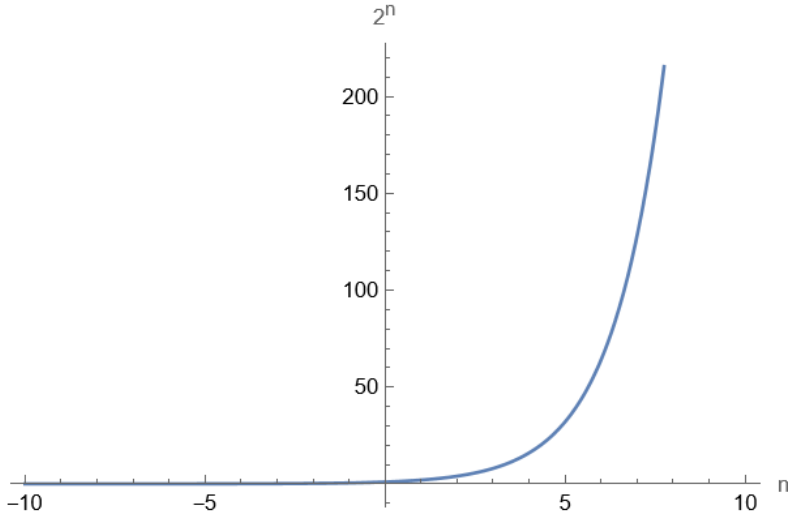
Exponential functions grow very quickly.



Example: Logarithms and exponentials

In general, we might write $g(n) = b^n$ or, equivalently, $g = \exp_b(n)$.

Exponential functions can be defined continuously, on $g : \mathbb{R} \to \mathbb{R}$.



Example: Logarithms and exponentials

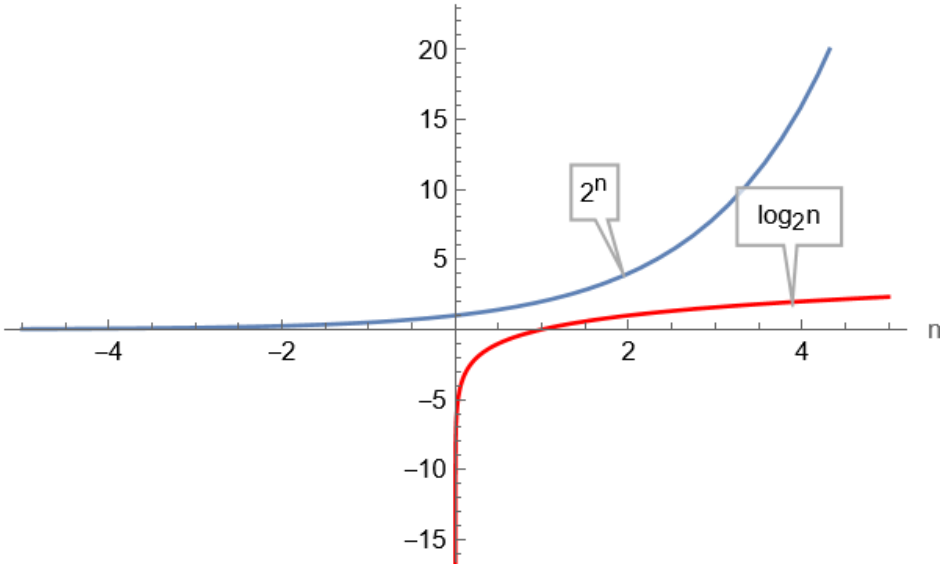
A *logarithm* is the inverse of an exponential function.

So, for $f(n) = 2^n$, we have

$$f^{-1}(n) = \log_2(n).$$

Hence,

$$\log_2(2^n) = n.$$



Example: Logarithms and exponentials

The most common exponentials and logarithms you will see are:

exponential	base	logarithm
2^n or $\exp_2 n$	2	$\log_2 n$
e^n or $\exp(n)$	e	$\ln(n)$ or $\log_e n$
10^n or $\exp_{10}(n)$	10	$\log_{10} n$ or just $\log n$

For a more in-depth explanation of logs and exponentials, you can attend the Academic Skills Centre workshop in Week 9.

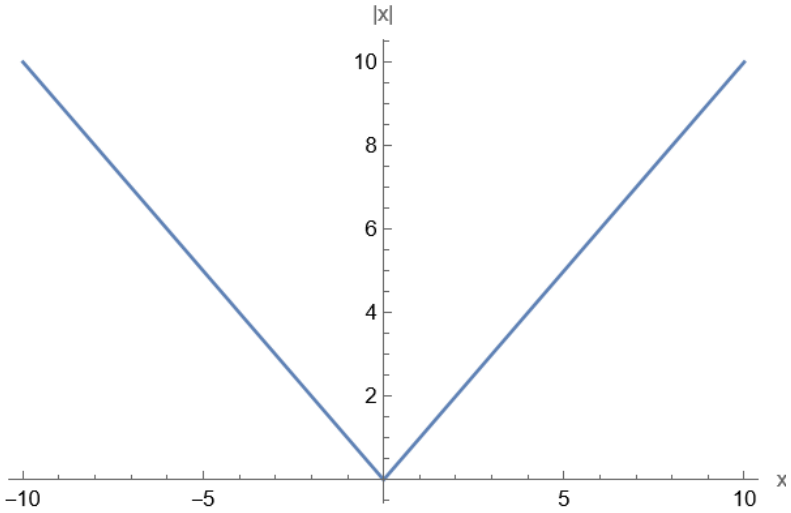
Since this workshop is so late in semester, you can also email me to request the slides from last semester's workshhop.

Piecewise functions

A *piecewise* function is defined differently on different subsets of its domain.

For example, the *absolute value* function, commonly written as $|x|$, or sometimes $\text{abs}\,(x)$, is defined as:

$$|x| = \begin{cases} -x & x < 0 \\ x & x \geq 0 \end{cases}$$



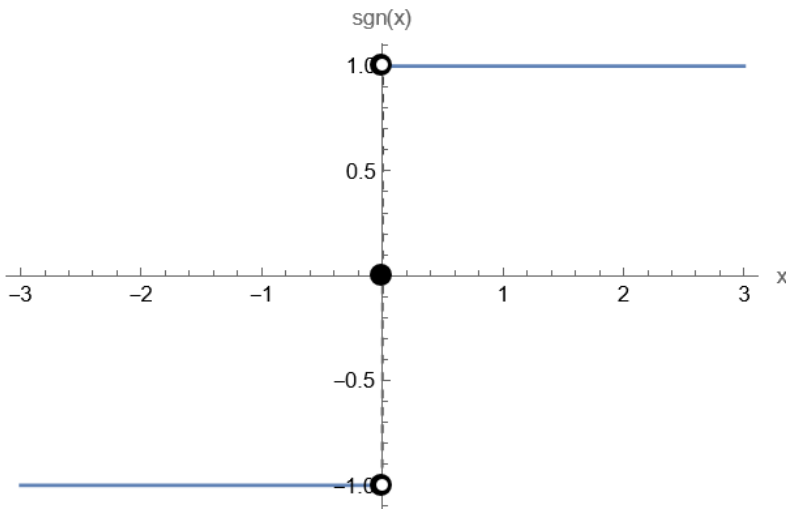
This function negates and inputs so that it only returns non-negative values.

Example: The sign function

A piecewise function you might have seen in the first CITS5508 lecture is the $sign$ function.

This fuction returns the sign of an input number:

$$\text{sgn}\,(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ +1 & x > 0 \end{cases}$$



Questions?

Is there anything from this session, or any maths you've seen in the first CITS5508 lecture, that you'd like to ask about?

Useful Resources

- Wikipedia's Glossary of Mathematical Symbols
- Academic Skills Centre online workshops:
  - Algebra (Week 2)
  - Summation (Week 3)
  - Logarithms and Exponentials (Week 9)
  - You can register for these workshops to receive the slides, even if you don't attend. If you missed a workshop, or want last semester's slides in advance, please email me to request a copy at jacey.ligthart-smith@uwa.edu.au

Academic Skills Centre: help with Maths & Stats

On-campus Drop-in: Maths, Stats & Programming
Tue, Wed, Thu, 10am-12pm *teaching weeks and study breaks.*
First Floor Reid Library

Build your skills with our advisors. Bring your friends & work together.

Consultations: Online or In-Person
Various times available, Monday to Friday
Click here for booking details

Build your skills with our advisors.

Workshops
View this Semester's topics and register for sessions here.

Want to know more? Got a question?
Send an email to jacey.ligthart-smith@uwa.edu.au

On-campus Drop-in: Writing & Study Skills
Tue, Wed, Thu, 10am-12pm *teaching weeks and study breaks.*
First Floor Reid Library

Improve your assignments with our advisors and student leaders

Online Consultations: Writing & Study Skills/English Language
Various times available, Monday to Friday
Click here for booking details

Improve your assignments with our advisors

Workshops
View this Semester's topics and register for sessions here.

Want to know more? Got a question?
Go to askUWA. Select "Academic Skills" from the "My enquiry is about" drop-down list.

The Academic Skills Centre Team

Using Academic Skills Centre Resources

We are happy for our resources to be used but please do not amend them, do not remove the "Academic Support webpage" link or UWA logos, and give credit/reference to the Academic Skills Centre team where necessary.

CC BY NC ND